

CONTINUATION

Helix OTA — Continuation / Resume Handoff

Field	Value
Revision	10
Created	2026-06-07
Last modified	2026-06-20T08:30:00Z
Status	active — resume with “continue”
Status summary	Single source of truth for resuming work. Captures exactly what is DONE (verified), the git state, and the prioritized NEXT steps. Everything below is committed to main and pushed to all upstreams (GitHub, GitLab, GitVerse). GitFlic rejects pushes (packfile truncated — tracked, non-blocking pending resolution).

↗ CURRENT STATE (2026-06-20) — HEAD 3e627daa — read docs/RESUMPTION.md FIRST

All four Stream-V parallel streams (M — code review PWU-AB-4, N — OpenCV vision validation, P — OTA Manager desktop recordings, U — §11.4.108 registry) merged into 3e627daa — the final merge batch.

Stream M — Independent code review PWU-AB-4 ApplyPort — DONE: - Code review agent analysed all PWU-AB-4 fixes (applyport, device client, slot manager) - Three findings fixed: server/cmd/applyport/main.go (6 insertions), server/internal/device/client.go (20 insertions, 5 deletions), server/internal/device/slot.go (3 insertions) - All fixes verified, committed, and merged as e210280c

Stream N — OpenCV-based vision validation for recordings — DONE: - Created scripts/testing/analyze_recording.sh + analyze_recording.py — automated OpenCV analysis pipeline that detects frozen frames, checks frame-advance, validates window dimensions - Batch 1 analysis on 34 MP4 recordings across all feature areas — 32 P / 0 F / 2 S (2 skipped — CLI interaction recordings

where prompt-wait-out produced terminal window) - Batch 3 analysis (re-recording run) on 32 recordings — 4 PASS, 28 FAIL (frozen/stale captures — most recordings were static single-frame clips, not genuine advancing content) - Batch 2 produced a transcript-based verification report (operator-side) - Created guide: docs/guides/video_recording_and_analysis.md - Evidence: docs/qa/20260620T063615Z-opencv-analysis/REPORT.md, docs/qa/20260620T074740Z-recording-analysis-batch3/REPORT.md - Merged as 63341246

Stream P — OTA Manager desktop app recordings — DONE: - Recorded OTA Manager desktop application with 8 window-scoped screenshots at 5s intervals - Final summary screenshot captured - Evidence at docs/qa/20260620T064840Z-ota-manager/REPORT.md - Merged as 37f480f8

Stream U — §11.4.108 runtime-signature registry for PWU-AB features — DONE: - Created docs/design/rk3588_ab_virt/runtime-signatures.yaml with 7 entries covering all 4 PWU-AB tiers - Each entry declares ONE machine-checkable runtime signature with real evidence paths, check commands, and determinism counts - Updated Issues.md (no change needed — PWU-AB items were already closed), Fixed.md (new OTA-015–OTA-018), Fixed_Summary.md - Merged as 5b885219; the final merge batch (3e627daa) folded all streams together

Previously landed PWU-AB milestones (HEAD 42be557): On the emulator A/B ladder (T1 = QEMU virt + HVF on this macOS host, real U-Boot 2024.01): **PWU-AB-1 FULL A/B slot switch is PROVEN** (evidence docs/qa/20260611T094958Z-ab-slot-switch/) **AND PWU-AB-3 corrupt-slot AUTO-ROLLBACK is PROVEN** (evidence docs/qa/20260611T095918Z-ab-rollback/). PWU-AB-2 RAUC dm-verity successfully reached GREEN in Stream A (merged earlier). **T2 Cuttlefish** real-Android-A/B remains **SKIP-pending** the operator's incoming Linux + nested-KVM host (/dev/kvm absent on this Apple-Silicon host); **T3 RK3588** hardware PENDING (no board). **No release tag** — §11.4.40 needs the full ladder GREEN (T2 + T3 still SKIP/PENDING), so a tag would be a bluff (§11.4.6). Everything below this box is prior-wave history.

Stream-V post-merge state

Recording corpus: 30 MP4 files at \$HOME/Downloads/helix_ota---*.mp4. Batch 1 (OpenCV) scored 32/34 PASS (mostly fake-window screensavers); Batch 3 (re-recording, 32 files) scored 4 PASS, 28 FAIL — the vast majority were frozen/stale single-frame captures lacking genuine advancing content. A remediation re-recording pass is pending to bring the corpus to §11.4.158 read-the-screen standards.

Push status: - GitHub: up-to-date - GitLab: up-to-date - GitVerse: up-to-date - **GitFlic: REJECTED** (packfile truncated — persists from earlier push failure gitlab_20260611T070926Z.log). Next push to GitFlic will require operator action (retry with --force-with-lease negotiated per §11.4.113 merge-onto-main path).

Pre-build verification: PASS (all gates including §11.4.156/§11.4.157/§11.4.158).

GEMINI.md lockstep: §11.4.158 anchor confirmed present (2 hits). Five-carrier family complete per §11.4.157.

Open items (Issues.md): | OTA-ID | § | Status | Type | Summary | |—|—|—|—|—| |
OTA-003 | §3 | Operator-blocked | Task | Tier-2 real Android A/B needs
Linux+KVM Cuttlefish host | | OTA-004 | §4 | Operator-blocked | Task | Tier-3
RK3588 hardware validation | | OTA-014 | §8 | In progress | Task | Docs Chain
submodule distribution (Phase 6) operator-gated |

features/Status.md: Rev 6, active at docs/features/Status.md — 73 feature rows documented across Server, Submodules, Dashboard, Emulator, Security, A/B categories. Video recording column populated per §11.4.153.

NEXT (immediate — risk-ordered per §11.4.132): 1. Recording remediation — re-record the 28 FAIL features with genuine advancing content (window-scoped per §11.4.154, prefixed per §11.4.155, content-read-verified per §11.4.158) 2. Retry GitFlic push when operator provides guidance on the packfile-truncated rejection 3. Unblock OTA-014 (Docs Chain Phase 6) on operator go-ahead

BLOCKED (hardware): OTA-003 T2 Cuttlefish (Linux+KVM host), OTA-004 T3 RK3588 board. No release tag until these unblock or an alternative §11.4.40-compliant gate is operator-approved.

Carried-forward gaps: All software-achievable tiers GREEN; remaining items hardware-gated or operator-gated (§11.4.21 self-resolution exhausted).